Realised Volatility Forecasting — Multi-Asset Benchmark

Direct Multi-Step Models: HAR, NLinear, DLinear, SAS, RNN

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1 Setup

$x_t = \log \mathrm{RV}_t$, z-scored per rolling window (μ, σ frozen until refit). Direct multi-step: one model per $h \in \{1, 5, 10\}$. Ridge $\alpha \in \{10^k : k \in [-4, 5],\ \Delta k{=}0.25\}$ (37 values), 5-fold CV.

2 Models

2.1 HAR — Heterogeneous Autoregressive Model (Corsi 2009)

Features. Let $\bar{x}_t^{(w)} = \frac{1}{5}\sum_{k=0}^{4} x_{t-k}$ and $\bar{x}_t^{(m)} = \frac{1}{22}\sum_{k=0}^{21} x_{t-k}$.

$$\hat{x}_{t+h} = \alpha + \beta_d\, x_t + \beta_w\, \bar{x}_t^{(w)} + \beta_m\, \bar{x}_t^{(m)} \tag{1}$$

HAR-OLS: $\min_\beta \|X\beta - y\|_2^2$. **HAR-Ridge**: column-normalise $[x^{(d)}, x^{(w)}, x^{(m)}]$; penalise slopes only; back-transform $w = w_{\text{scaled}}/\sigma_{\text{col}}$.

2.2 NLinear and DLinear (Zeng et al. 2023)

Let $\boldsymbol{x}_t = (x_{t-L+1}, \ldots, x_t)^\top \in \mathbb{R}^L$ be the lookback window.

NLinear.

$$\tilde{\boldsymbol{x}}_t = \boldsymbol{x}_t - x_t, \qquad \hat{x}_{t+h} = \tilde{\boldsymbol{x}}_t^\top w_h + x_t \tag{2}$$

DLinear.

$$\boldsymbol{\tau}_t = \mathrm{MA}_k(\boldsymbol{x}_t), \quad \boldsymbol{\varsigma}_t = \boldsymbol{x}_t - \boldsymbol{\tau}_t, \qquad \hat{x}_{t+h} = \begin{bmatrix} \boldsymbol{\tau}_t \\ \boldsymbol{\varsigma}_t \end{bmatrix}^\top w_h \tag{3}$$

2.3 GARCH — AR(p)-GARCH(1,1)

$x_t = \mu + \sum_{i=1}^{p} \phi_i x_{t-i} + \varepsilon_t, \quad \sigma_t^2 = \omega + \alpha_G \varepsilon_{t-1}^2 + \beta_G \sigma_{t-1}^2$. Multi-step: iterate variance recursion analytically. QMLE.

2.4 SAS — State Affine System

Diagonal basis. $P_0, \ldots, P_p \in \mathbb{R}^n,\ Q_0, \ldots, Q_q \in \mathbb{R}^n,\ z \in \mathbb{R}$.

$$P(z) = \mathrm{clip}\Big(\sum_{d=0}^{p} P_d\, z^d,\ -1{+}\epsilon,\ 1{-}\epsilon\Big) \in (-1,1)^n \tag{4}$$

$$Q(z) = \sum_{d=0}^{q} Q_d\, z^d \in \mathbb{R}^n \tag{5}$$

$$s_t = P(z_t) \odot s_{t-1} + Q(z_t) \tag{6}$$

Clipping ensures $|P(z)_i| < 1$ (unconditional stability). Readout: $\hat{x}_{t+h} = s_t^\top w_h$, W_h fitted by ridge regression with 5-fold CV.

Transition polynomial P. Let ρ be the spectral-norm bound and $\delta = \max(0.005,\ (1{-}\rho)\cdot 0.3)$ a small headroom margin.

- $P_0[i] \overset{\text{iid}}{\sim} \mathcal{U}(-\rho,\ \rho)$ — base eigenvalues spread uniformly to encode timescales from fast-decaying to near-unit-root.
- $P_k[i] \overset{\text{iid}}{\sim} \mathcal{N}\big(0,\ \delta/5^k\big)$ for $k \geq 1$ — exponentially shrinking input-modulation terms so that the clip boundary is rarely reached for $|z| \leq 5$.

Input-drive polynomial Q. The injection $Q(z_t)$ adds noise with variance $\propto \sum_k Q_k[i]^2\, \mathbb{E}[z^{2k}]$. For $z \sim \mathcal{N}(0,1)$, $\mathbb{E}[z^{2k}] = (2k-1)!! = 1\cdot 3\cdot 5\cdots(2k{-}1)$ (double factorial). To keep injection variance roughly uniform across degrees:

$$Q_k[i] \overset{\text{iid}}{\sim} \mathcal{N}\left(0,\ \frac{1}{\sqrt{n\,(2k-1)!!}}\right) \tag{7}$$

so $k{=}0,1$ use scale $1/\sqrt{n}$ (unchanged from a linear SAS), while $k{=}2$ is shrunk by $1/\sqrt{3} \approx 0.58$ and $k{=}3$ by $1/\sqrt{15} \approx 0.26$. This prevents state explosion for large inputs without hand-tuning per-degree scales.

Associative structure. $\ell_t = (P(z_t), Q(z_t))$. Composition rule:

$$(A_j,\ b_j) \circ (A_i,\ b_i) \ =\ \big(A_j \odot A_i,\ \ A_j \odot b_i + b_j\big) \tag{8}$$

is associative $\Rightarrow$ parallel divide-and-conquer scan.

Algorithm 1: Two-Level Parallel Scan — compute $s_1, \ldots, s_T$ from $z_1, \ldots, z_T$

Input : Transitions $\ell_1, \ldots, \ell_T$;;
chunk size B;;
$s_0 = \mathbf{0}$
Output : States $s_1, \ldots, s_T$;;
s_T

$K \leftarrow \lceil T/B \rceil$; pad to KB
Phase 1 (vmap over K chunks);
  $\forall\, k:\ (A^{\text{intra}}_{k,\cdot},\ b^{\text{intra}}_{k,\cdot}) \leftarrow \textsc{AssocScan}(\circ,\ \ell_{kB+1}, \ldots, \ell_{(k+1)B})$
Phase 2 (inter-chunk scan over K endpoints);
  $(A^{\text{inter}},\ b^{\text{inter}}) \leftarrow \textsc{AssocScan}(\circ,\ (A^{\text{intra}}_{1,B}, b^{\text{intra}}_{1,B}), \ldots)$
Phase 3 (carry for chunk $k \geq 1$);
  $c_0 = s_0$;;
$c_k = A^{\text{inter}}_{k-1} \odot s_0 + b^{\text{inter}}_{k-1}$
Phase 4 (vmap over K chunks);
  $\forall\, k, j:\ s_{kB+j} = A^{\text{intra}}_{k,j} \odot c_k + b^{\text{intra}}_{k,j}$

Depth $O(\log T)$ vs. $O(T)$ sequential.

2.5 RNN — LSTM / GRU / LRU

Readout: $\hat{x}_{t+h} = \text{Linear}_h(h_T)$, one head per h. Training: Adam, early stopping (patience=20, val=20% chronological).

LRU (Orvieto et al. 2023):

$$h_t = \sigma(\nu) \odot h_{t-1} + Bz_t, \qquad y_t = Ch_t, \qquad \sigma(\nu) \in (0,1)^d \tag{9}$$

3 Evaluation Protocol

Algorithm 2: Rolling Out-of-Sample Evaluation

Input : Series $x_1, \ldots, x_T$; window W; refit frequency R; horizons $\mathcal{H}$
Output : Squared-error sequences $\{e_{t,m}^{(h)}\}$ for each model m, horizon h

```
(μ, σ) ← (0, 1);
for t = W to T − max(H) do
    train ← x_{t−W:t};
    if t mod R = 0 then
        (μ, σ) ← FitScaler(train);
        traiñ ← (train − μ)/σ;
        foreach model m do
            m.Fit(traiñ, H)
    z_t ← (x_t − μ)/σ;
    foreach model m, horizon h ∈ H do
        ẑ_{t+h} ← m.Predict(h);
        e^{(h)}_{t,m} ← (ẑ_{t+h} − z_{t+h})^2;
    foreach model m do
        m.Update(z_t)
```

Model Confidence Set (Hansen et al. 2011). $L \in \mathbb{R}^{T \times M}$, row-normalised by cross-model mean to prevent high-vol dominance.

1. $d_{t,m} = L_{t,m} - \bar{L}_t$
2. $T_R = \max_m \bar{d}_m / \hat{\sigma}_m$ ($\hat{\sigma}_m$: circular block bootstrap, B reps)
3. Eliminate worst model if $p < \alpha$; repeat until H_0 not rejected.
4. Surviving set = MCS_α.

MCS Frequency. Run MCS per symbol per horizon; report $c_m^{(h)}/N$.

4 Hyperparameters

Table 1: Fixed hyperparameters used in all experiments.

Component	**Parameter**	**Value**
Evaluation	Training window W	2 000
	Refit frequency R	20 steps ($\approx$1 month)
	Horizons $\mathcal{H}$	$\{1, 5, 10\}$ days
MCS	Significance α	0.10
	Bootstrap replications B	1 000
HAR	Lags (d, w, m)	$(1, 5, 22)$ days
	Ridge CV folds	5
NLinear / DLinear	Lookback L	20
	DLinear MA kernel	5
GARCH	AR lags p	5
	Estimation	QMLE
SAS (diagonal)	Reservoir size n	200
	Spectral norm ρ	0.95
	P-polynomial degree p	1
	Q-polynomial degree q	1 ($\mathrm{SAS_{q1}}$) or 2 ($\mathrm{SAS_{q2}}$)
	Washout	50 steps
RNN (GRU)	Hidden size	32
	Layers	1
	Lookback	20
	Epochs (max)	500
	Early stopping	patience = 20, val = 20%
MdSAS	Groups (D+1)	2 (degrees 0,1) or 3 (degrees 0,1,2)
	n per group	50 or 100
	Spectral norm ρ	$\{0.90, 0.95\}$
	α grid (grouped)	7-value subset (Δk=1.5, 49 combos)
Ridge grid	Range	10^k, $k \in [-4, 5]$
	Spacing (single-α)	$\Delta k = 0.25$, 37 values
	Spacing (grouped)	$\Delta k = 1.5$, 7 values (49 combos/2 groups)

5 Results

5.1 All Models — 30 Symbols, $h \in \{1, 5, 10\}$

Table 2: Out-of-sample MSE — mean ± std across 30 symbols (W=2000, R=20, target: $\log \mathrm{RV}_5$, N_{sym}=30). Sorted by avg MSE. Bold: lowest mean per horizon. $\mathrm{SAS}_{\mathrm{q1/q2}}$: diagonal reservoir, n=200, ρ=0.95, p=1.

Model	$h = 1$	$h = 5$	$h = 10$	avg
$\mathrm{SAS}_{\mathrm{q2}}$	$\mathbf{0.3958 \pm 0.0718}$	0.5070 ± 0.0934	0.5837 ± 0.1145	**0.4955**
HAR	0.4002 ± 0.0721	0.5090 ± 0.0928	$\mathbf{0.5827 \pm 0.1053}$	0.4973
$\mathrm{SAS}_{\mathrm{q1}}$	0.3973 ± 0.0713	$\mathbf{0.5065 \pm 0.0913}$	0.5914 ± 0.1058	0.4984
DLinear	0.4001 ± 0.0720	0.5116 ± 0.0927	0.5855 ± 0.1057	0.4991
NLinear	0.4032 ± 0.0741	0.5191 ± 0.0973	0.5977 ± 0.1130	0.5067
GRU	0.4200 ± 0.0848	0.5410 ± 0.1107	0.6249 ± 0.1323	0.5287
GARCH	0.4104 ± 0.0741	0.5459 ± 0.0976	0.6466 ± 0.1182	0.5343

Table 3: Model Confidence Set (MCS, α=0.10, B=1000 bootstrap replications, Hansen et al. 2011) — frequency of inclusion in the superior set across 30 symbols, sorted by total. A model survives if it cannot be shown to be significantly worse than the best model on that symbol. NLinear is excluded (0 survivals across all horizons).

Model	$h = 1$	$h = 5$	$h = 10$	total	avg MSE
HAR	15/30 (50%)	29/30 (97%)	30/30 (100%)	74/90	0.4973
$\mathrm{SAS}_{\mathrm{q1}}$	29/30 (97%)	24/30 (80%)	19/30 (63%)	72/90	0.4984
$\mathrm{SAS}_{\mathrm{q2}}$	18/30 (60%)	14/30 (47%)	9/30 (30%)	41/90	0.4955
DLinear	12/30 (40%)	11/30 (37%)	14/30 (47%)	37/90	0.4991
GRU	0/30 (0%)	2/30 (7%)	1/30 (3%)	3/90	0.5287
GARCH	1/30 (3%)	0/30 (0%)	0/30 (0%)	1/90	0.5343

5.2 SAS as Data-Generating Process — Specification Experiment

Algorithm 3: Extended SAS-DGP Specification Experiment

Input : Set of real-data symbols $\mathcal{Y}$;;
Noise distribution $\mathcal{D}$;;
Number of steps N_{gen};;
Set of seeds $\mathcal{M} = \{s_1, s_2, s_3\}$
Output : Synthetic series $Z^{(y,m)}$ for each symbol $y \in \mathcal{Y}$ and seed $m \in \mathcal{M}$

for *each symbol* $y \in \mathcal{Y}$ **do**
  `// Step 1: Fit SASq2 on full real series`
  Fit $\text{SAS}_{q=2}$ on the full observation sequence $y_{1:T}$;
  Extract reservoir matrices (P, Q), readout W_1, terminal state s_T, and noise scale $\hat{\sigma}$;
  for *each seed* $m \in \mathcal{M}$ **do**
    `// Step 2: Autoregressive generation from terminal state`
    Set random number generator seed to m;
    $s \leftarrow s_T$;
    Initialize empty sequence $Z^{(y,m)}$;
    for $t = 1$ **to** N_{gen} **do**
      $\varepsilon \sim \mathcal{D}$;
      $z_t \leftarrow s^\top W_1 + \hat{\sigma}\,\varepsilon$;
      $s \leftarrow P(z_t) \odot s + Q(z_t)$;
      Append z_t to $Z^{(y,m)}$;

Table 4: DGP specification experiment — Heavy-tailed noise. DGP: SAS_{q2} (p=1, n=50, ρ=0.95), noise $\mathcal{D} = t(5)$, no clipping, N_{gen}=4,000. Evaluated over N=56 symbol-seed runs. SAS DGP is the exact oracle floor ($\text{MSE}_{\text{oracle}} = \hat{\sigma}^2 \cdot \nu/(\nu{-}2)$). MCS uses α=0.10 and B=1000 bootstraps, with losses row-normalised.

Model	MSE ($h = 1$)	MCS Survival	MCS %
SAS DGP (oracle)	$\mathbf{0.5084 \pm 0.1207}$	30/56	54%
`SAS_same_fixed`λ	0.5127 ± 0.1228	**49/56**	**88%**
`SAS_same_cv`λ	0.5129 ± 0.1223	48/56	86%
`SAS_diff`	0.5173 ± 0.1312	28/56	50%
`SAS_q1`	0.5213 ± 0.1285	41/56	73%
HAR	0.5271 ± 0.1436	2/56	4%
DLinear	0.5286 ± 0.1449	1/56	2%
GARCH	0.5327 ± 0.1423	0/56	0%

Table 5: DGP specification experiment — Gaussian noise. DGP: SAS_{q2} (p=1, n=50, ρ=0.95), noise $\mathcal{D} = \mathcal{N}(0, 1)$, no clipping, N_{gen}=4,000. Evaluated over N=69 symbol-seed runs. SAS DGP is the exact oracle floor ($\text{MSE}_{\text{oracle}} = \hat{\sigma}^2$). MCS uses α=0.10 and B=1000 bootstraps, with losses row-normalised.

Model	MSE ($h = 1$)	MCS Survival	MCS %
SAS DGP (oracle)	$\mathbf{0.3015 \pm 0.0678}$	**65/69**	**94%**
`SAS_same_fixed`λ	0.3898 ± 0.0832	10/69	14%
`SAS_diff`	0.3906 ± 0.0833	4/69	6%
`SAS_same_cv`λ	0.3913 ± 0.0820	9/69	13%
HAR	0.3917 ± 0.0842	1/69	1%
DLinear	0.3920 ± 0.0839	0/69	0%
GARCH	0.4005 ± 0.0870	0/69	0%
`SAS_q1`	0.4810 ± 0.7447	10/69	14%

5.3 Polynomial Degree Sweep (p and q)

Table 6: Polynomial degree sweep — Mean Squared Error. Evaluated across all N=30 symbols. Means for $p = 1, q = 0$ are heavily inflated by divergences. All SAS models use n=200 and spectral norm limit ρ=0.95.

Model	$h = 1$	$h = 5$	$h = 10$	avg
SAS (p=1, q=2)	0.3400 ± 0.0632	0.4860 ± 0.0896	0.5713 ± 0.1141	0.4658
HAR	0.3427 ± 0.0628	0.4856 ± 0.0869	0.5695 ± 0.1044	0.4659
SAS (p=1, q=1)	0.3410 ± 0.0618	0.4843 ± 0.0858	0.5783 ± 0.1045	0.4679
SAS (p=0, q=1)	0.3404 ± 0.0621	0.4965 ± 0.0896	0.5687 ± 0.1050	0.4685
SAS (p=0, q=2)	0.3400 ± 0.0632	0.4944 ± 0.0970	0.5759 ± 0.1156	0.4701
SAS (p=1, q=0)	0.4937 ± 0.3138	2.1042 ± 4.8994	0.7790 ± 0.5379	1.1257

Table 7: Polynomial degree sweep — MCS frequency. Evaluated on the stable subset of N=26 symbols. Losses are row-normalised. α=0.10, B=1000 bootstrap replications. Sorted by total survivals.

Model	$h = 1$	$h = 5$	$h = 10$	Total Survivals	avg MSE
SAS (p=1, q=1)	26/26 (100%)	25/26 (96%)	25/26 (96%)	76	0.4683
SAS (p=0, q=1)	23/26 (88%)	21/26 (81%)	20/26 (77%)	64	0.4688
SAS (p=1, q=2)	18/26 (69%)	17/26 (65%)	15/26 (58%)	50	0.4659
SAS (p=0, q=2)	19/26 (73%)	16/26 (62%)	13/26 (50%)	48	0.4681
SAS (p=1, q=0)	0/26 (0%)	3/26 (12%)	3/26 (12%)	6	0.6570
HAR	2/26 (8%)	0/26 (0%)	0/26 (0%)	2	0.4659